R2R: Reasoning-to-Residual Alignment for Financial Time Series Forecasting via Fine-Tuning LLMs

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Abstract

Financial time series forecasting remains a formidable challenge due to the disconnect between endogenous historical trends and exogenous macroeconomic shocks. While state-of-the-art channel-independent transformers excel at capturing long-term temporal dependencies, they remain semantically blind to regime-shifting events like monetary policy adjustments. Existing multimodal approaches often fail to bridge this gap, as naive fusion mechanisms frequently lead to modality interference, where high-dimensional textual noise obscures the underlying technical signals. To address this challenge, we propose a novel Reasoning-to-Residual Alignment framework that reformulates the integration of economic logic as a residual learning task. Our framework decouples the forecasting process into two components: a technical backbone that generates a baseline numerical trajectory, and the LLM that functions as a specialized residual predictor. Crucially, the LLM is fine-tuned not on raw price generation, but on regressing the forecast residual (i.e., the residual between the baseline and ground truth) conditioned on macro-event textual data. Such alignment enables the LLM to explicitly learn the quantitative mapping from semantic news to the volatility corrections required by the technical model. The final forecast is derived by rectifying the baseline momentum with the LLM-predicted residual. Extensive experiments on financial datasets demonstrate that our approach significantly reduces Mean Squared Error (MSE) compared to vanilla baselines, proving that fine-tuned semantic models can effectively calibrate the blind spots of numerical forecasters.

Keywords

Time Series Forecasting, Large Language Models, Fine-tuning, Financial reasoning, Residual Alignment

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1 Introduction

Financial time series forecasting remains one of the most intractable challenges in data mining, primarily due to the stochastic interplay between two fundamentally distinct generative processes: (1) Endogenous Microstructure, characterized by high-frequency continuous patterns such as momentum and mean reversion [3, 21, 25, 28]; and (2) Exogenous Macro-Dynamics, driven by discrete, regime-shifting policy events (e.g., FOMC decisions, CPI releases) [10, 15]. While endogenous patterns obey historical statistical laws, exogenous shocks often introduce non-stationary structural breaks that defy purely numerical extrapolation [13, 14, 20].

In recent years, the field has witnessed a paradigm shift with the advent of channel-independent transformers, such as PatchTST [15] and iTransformer [10]. These methods adopt a design choice that processes each variable as a separate channel and have shown strong empirical performance on long-horizon prediction. Meanwhile, a parallel line of work explores leveraging LLMs for time series via reprogramming or autoregressive prompting [6, 11, 16], while also examining when the LLM component is the true source of the observed gains [12, 19, 24]. However, despite their architectural sophistication, they essentially operate as closed systems. Relying solely on historical numerical windows, they inherently assume distributional continuity. Consequently, when an exogenous shock triggers a market regime shift [18], the semantic information driving the shift remains invisible to their numerical encoders, leading to significant concept drift [5, 8, 10, 25].

To incorporate semantic information, recent advancements [6, 11, 26] have attempted to integrate LLMs into forecasting, generally falling into two paradigms: direct generation and embedding fusion. However, both approaches face critical bottlenecks:

(1) **Arithmetic Hallucination in Direct Generation:** Methods like Time-LLM [6] repurpose LLMs as numerical generators via tokenization. However, mapping continuous financial values to discrete linguistic tokens lacks fundamental arithmetic grounding. Consequently, these probabilistic models often fail to capture precise regression dynamics, leading to hallucinations that violate market microstructure constraints.
(2) **Modality Misalignment in Embedding Fusion:** Multimodal baselines typically employ naive concatenation, which introduces a dual disconnect. First, the semantic-numerical gap: generic NLP embeddings capture linguistic proximity rather than economic magnitude (e.g., differentiating dovish vs. hawkish sentiment). Second, the impedance mismatch: forcing sparse, high-dimensional text vectors into dense time-series models introduces significant noise, obscuring dominant technical trends during non-event periods.

To resolve these challenges, we propose the **R2R** (Reasoning-to-Residual) framework. Our approach draws inspiration from the cognitive workflow of professional traders, who typically establish a baseline expectation based on historical momentum but apply discrete corrections upon receiving critical policy news. Guided by this intuition, our main contributions are as follows:

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- **Reasoning-to-Residual Alignment.** We propose R2R, a novel paradigm that fundamentally reframes financial forecasting as the superposition of two orthogonal components: the endogenous prior and the exogenous residual. Specifically, we utilize a standard time series backbone (e.g., GPT4TS [30]) to capture the market's inertia within a closed numerical system, producing the endogenous prior, which is a baseline trajectory independent of external shocks. Concurrently, we reframe the LLM's role from a passive text encoder to an active semantic calibrator. By mathematically bridging these components via multiplicative modulation, we effectively mitigate the semantic blindness of traditional transformers, preserving technical fidelity during quiet periods while aggressively rectifying trajectory shifts during macro-regime changes.
- **Exogenous Residual Construction and Denoising.** Instead of forcing the LLM to hallucinate absolute prices, we fundamentally reframe the forecasting objective from absolute price generation to semantic residual prediction, positing that linguistic signals correlate strictly with trend deviations rather than market inertia. Crucially, we challenge the reliance on raw ground-truth residuals, which are often polluted by high-entropy microstructure noise. Instead, we introduce a novel LLM-guided denoising mechanism to distill raw residuals into smoothed, logically-grounded vectors. By replacing noisy ground truth with these denoised proxies, we ensure the model internalizes deterministic macro-economic logic rather than overfitting to stochastic market volatility.
- **Empirical Validation and Reproducibility:** We evaluate on hourly financial datasets across multiple assets, including BTC, ETH, DOGE, SOLANA, SPX, NDX, and INDU. The proposed framework outperforms strong baselines and recent multimodal methods on the vast majority of cases, showing robust performance under high-volatility and policy-event regimes. Notably, during the *Fed Chair Powell Testifies* event, our method achieves substantial MAE reductions (up to 75.55%). The framework also delivers superior risk-adjusted trading performance and competitive computational efficiency. Code will be released upon publication (https://anonymous.4open.science/r/R2R-6636).

2 Related Work

Time series forecasting has evolved from purely numerical modeling toward incorporating external semantic signals.

Time Series Numerical Modeling: The landscape of time series forecasting has evolved from neural networks to sophisticated transformer architectures. Early efforts like LogTrans [7] and Informer [28] introduced sparse attention mechanisms to mitigate computational complexity. Subsequently, decomposition-based models such as Autoformer [22] and Fedformer [29] integrated trend-seasonality blocks to handle complex non-stationary data.

Current approaches emphasize channel independence and patching mechanisms. DLinear [25] challenges transformer dominance by employing a simple linear mapping with decomposition. PatchTST [15] further advances this by segmenting time series into sub-series patches. Similarly, GPT4TS [30] focuses on numerical pattern matching and remains semantically blind to external narrative shocks. iTransformer [10] inverts the dimensions of the attention mechanism to capture multivariate correlations. However, they rely exclusively on historical numerical patterns and lack the semantic sensors required to detect exogenous policy shocks.

Incorporating Semantic Signals via LLMs: The emergence of LLMs has sparked interest in integrating textual data into time series forecasting. Early approaches adopted a late fusion strategy, concatenating BERT-based embeddings with numerical features. However, standard embeddings often struggle to capture domain-specific economic logic.

More recent works attempt to leverage LLMs directly for forecasting. Time-LLM [6] proposes reprogramming LLMs by quantizing time series into text tokens, effectively treating forecasting as a next-token prediction task. UniTime [9] attempts to build a foundation model for cross-domain time series using language prompts. TimesNet [21] introduces a multi-periodicity analysis framework, serving as a robust general-purpose encoder. TEST [17] proposes aligning time series embeddings with text prototypes to activate the latent capabilities of LLMs. Most recently, CAMEF [27] integrates textual news and numerical price series via a neural network for joint forecasting. Despite these advances, implicit fusion strategies struggle to disentangle endogenous momentum from exogenous shocks, lacking a quantitative mechanism to rectify policy-induced numerical deviations.

3 Preliminaries

Financial markets are driven by the discrete arrival of macroeconomic information, which often triggers structural breaks that technical models cannot foresee. We categorize the high-impact exogenous shock events $\mathcal{E}$ in our study into four semantic pillars:

- **Monetary Policy:** This includes FOMC rate decisions, statements, and Fed chair speeches. These events dictate the cost of capital and liquidity regimes [4]. The semantic nuance (e.g., Hawkish vs. Dovish) directly shifts market expectations.
- **Inflationary Pressure:** Indicators such as CPI, PPI, and Core PCE quantify the purchasing power and future policy constraints. Unexpected inflation typically induces immediate price corrections [1].
- **Labor Market Health:** Monthly Nonfarm Payrolls (NFP), unemployment rates, and weekly jobless claims serve as proxies for economic resilience. Labor data is a primary determinant of central bank intervention thresholds.
- **Real Economy Activity:** This pillar encompasses ISM Manufacturing/Services PMI, GDP growth, and Industrial Inventories (e.g., Crude Oil). These metrics reflect the fundamental growth trajectory and supply-demand balance of the economy.

4 Overview

The **R2R (Reasoning-to-Residual Alignment)** framework decouples financial forecasting into neural market perception and reasoning-driven shock calibration (Fig. 1). The pipeline consists of four stages:

Stage 1: Endogenous Prior. A technical backbone Φ_{base} produces the neural forecast $\hat{\mathbf{Y}}^{\text{base}}_{t:t+H}$ and estimates the pre-event technical bias b_t, capturing systematic drift before macro shocks.

Stage 2: Exogenous Residual Construction. We isolate policy-driven signals by comparing ground truth with the technical baseline. A Teacher LLM performs denoising to convert noisy residuals into structured correction vectors $\mathbf{R}_t^*$ with reasoning traces, forming a JSON target $\mathcal{S}_t$.

Stage 3: Structured Residual Learning. A student model Ψ_{SFT} is fine-tuned via LoRA to map macro narratives and technical bias into quantitative residual corrections through structured generation.

Stage 4: Synergistic Calibration. During inference, Φ_{base} produces the neural prior while Ψ_{SFT} generates residual corrections $\hat{\mathbf{R}}_t^*$. The final forecast is obtained via multiplicative fusion, enabling dynamic calibration only under significant macro shocks.

5 Methodology

As shown in Algorithm 1, the core philosophy of the R2R framework is the orthogonal decomposition of market dynamics. We posit that financial time series are not monolithic but are composed of endogenous momentum and exogenous structural shocks.

5.1 Problem Formulation

We formulate event-driven financial forecasting as a dual-component problem, aiming to modulate endogenous historical trends with exogenous semantic shocks.

Let the continuous market state be denoted as $\mathbf{X}_{t-L:t} \in \mathbb{R}^{L\times V}$, representing the multivariate numerical history (e.g., price, volume) over a look-back window L. Traditional technical analysis relies on a baseline patch times series encoder Φ_{base} (e.g., PatchTST, GPT4TS) to extract temporal dependencies and generate a forecast trajectory over a future horizon H. We designate this baseline as the endogenous prior, denoted as $\hat{\mathbf{Y}}^{\text{base}}_{t:t+H}$, because it relies exclusively on internal market signals such as historical price and volume trends. Although this technical encoder is proficient at capturing numerical patterns, it is fundamentally semantically blind and lacks the capacity to interpret unstructured textual data. Since Φ_{base} operates under the rigid assumption of historical continuity, it remains incapable of anticipating the abrupt regime shifts that occur when exogenous policy shocks disrupt the market equilibrium.

Concurrently, let $\mathcal{E}_t$ denote the stream of unstructured exogenous events (e.g., policy announcements) available at time t. To bridge the gap between numerical trends and semantic shocks, we introduce a reasoning-to-residual mechanism (see Algorithm 1). Instead of predicting absolute prices via LLMs, we fine-tune an LLM, denoted as Ψ_{SFT}, to interpret $\mathcal{E}_t$ and predict a semantic residual vector of length H.

Crucially, we model the impact of news not as an additive error, but as a *multiplicative modulation*. We use a multiplicative form because the base model predicts the trajectory, and news mainly strengthens or weakens that trajectory rather than shifting it by a fixed amount. This keeps the original trend while letting news adjust how strong the movement is. The final forecast $\hat{\mathbf{Y}}_{t:t+H}$ is synthesized via leakage-free calibration:

$$\hat{\mathbf{Y}}_{t:t+H} = \underbrace{\hat{\mathbf{Y}}^{\text{base}}_{t:t+H}}_{\text{Endogenous Prior}} \odot (\mathbf{1} + \underbrace{\Psi_{\text{SFT}}(\mathcal{E}_t)}_{\text{Semantic Residual}}) \tag{1}$$

Algorithm 1 R2R: Reasoning-to-Residual Alignment

Require: Historical time-series $\mathbf{X}$; Ground truth trajectories $\mathbf{Y}$; Macro-policy event stream $\mathcal{E}$; Teacher LLM $\mathcal{M}_{\text{teach}}$.
Ensure: Calibrated forecast $\hat{\mathbf{Y}}_{t:t+H}$.

Stage 1: Endogenous Prior: Neural Perception
Train technical backbone Φ_{base} on $\mathbf{X}$ to minimize MSE loss
Initialize training dataset $\mathcal{D}_{\text{SFT}} \leftarrow \emptyset$
for each training instance t **do**
  $\hat{\mathbf{Y}}^{\text{base}}_{t:t+H} \leftarrow \Phi_{\text{base}}(\mathbf{X}_{t-L:t})$ ▷ Endogenous prior
  $b_t \leftarrow \text{Mean}(\mathbf{Y}_{t-k:t} \oslash \hat{\mathbf{Y}}^{\text{base}}_{t-k:t} - \mathbf{1})$ ▷ Ex-ante price bias
end for
Stage 2: Exogenous Residual Construction and Teacher LLM-Guided Denoising
for each training instance t **do**
  $\mathbf{R}_{\text{raw}} \leftarrow (\mathbf{Y}_{t:t+H} \oslash \hat{\mathbf{Y}}^{\text{base}}_{t:t+H}) - \mathbf{1}$ ▷ Raw noisy residual
  $(\mathbf{R}_t^*, \mathcal{A}_t) \leftarrow \mathcal{M}_{\text{teach}}(\mathbf{R}_{\text{raw}}, \mathcal{E}_t, b_t \mid \mathcal{P}_{\text{denoise}})$
  $\mathcal{S}_t \leftarrow \{\text{reasoning} : \mathcal{A}_t, \text{vector} : \mathbf{R}_t^*\}$ ▷ Structured target
  $\mathcal{D}_{\text{SFT}} \leftarrow \mathcal{D}_{\text{SFT}} \cup \{(\mathcal{P}, \mathcal{E}_t \oplus \mathcal{A}_t \oplus b_t, \mathcal{S}_t)\}$
end for
Stage 3: Fine-Tuning Exogenous Residual via Structured Generation
Initialize Ψ_{SFT} with pre-trained $\mathcal{M}_{\text{student}}$ and LoRA adapters
Minimize $\mathcal{L}_{\text{SFT}}(\theta) = -\sum_j \log P_\theta(y_j \mid \mathcal{P}, \mathcal{C}_\tau, y_{<j})$ ▷ Causal Language Modeling
Stage 4: Synergistic Inference and Calibration
$\mathcal{C}_t \leftarrow \mathcal{E}_t \oplus \mathcal{A}_t \oplus b_t$ ▷ Synthesize hybrid input context
$\hat{\mathcal{S}}_t \leftarrow \Psi_{\text{SFT}}(\mathcal{P}, \mathcal{C}_t)$ ▷ Structured JSON response autoregressively
$\hat{\mathbf{R}}_t^* \leftarrow f_{\text{ext}}(\hat{\mathcal{S}}_t.\texttt{vector})$ ▷ Semantic Residual via parsing operator
$\hat{\mathbf{Y}}_{t:t+H} \leftarrow \hat{\mathbf{Y}}^{\text{base}}_{t:t+H} \odot (\mathbf{1} + \hat{\mathbf{R}}_t^*)$ ▷ Multiplicative fusion
return $\hat{\mathbf{Y}}_{t:t+H}$

where $\odot$ denotes the element-wise Hadamard product. In this formulation, Ψ_{SFT} outputs a vector in $\mathbb{R}^H$ representing the relative residual ratio for each future time step.

5.2 Stage 1: Endogenous Prior: Neural Perception

Given the current look-back window $\mathbf{X}_{t-L:t}$, the technical backbone generates the baseline trajectory. We utilize GPT4TS [30] as the encoder Φ_{base}. By segmenting the multivariate series into overlapping patches, GPT4TS aggregates local temporal information, mitigating the high-frequency noise inherent in financial markets. The model is trained using a Mean Squared Error (MSE) loss on purely numerical historical data. We define the output $\hat{\mathbf{Y}}^{\text{base}}_{t:t+H}$ as the endogenous prior, representing the price path the market would follow if driven solely by historical microstructure mechanics:

$$\hat{\mathbf{Y}}^{\text{base}}_{t:t+H} = \Phi_{\text{base}}(\mathbf{X}_{t-L:t}) \tag{2}$$

5.3 Stage 2: Exogenous Residual Construction and Teacher LLM-Guided Denoising

A critical challenge in training the residual predictor Ψ_{SFT} is the stochastic nature of financial markets. The raw forecast residual between the ground truth $\mathbf{Y}$ and the baseline $\hat{\mathbf{Y}}_{base}$ contains not only the *deterministic signal* induced by the macro event but also *aleatoric noise* from market microstructure (e.g., liquidity vacuums or random

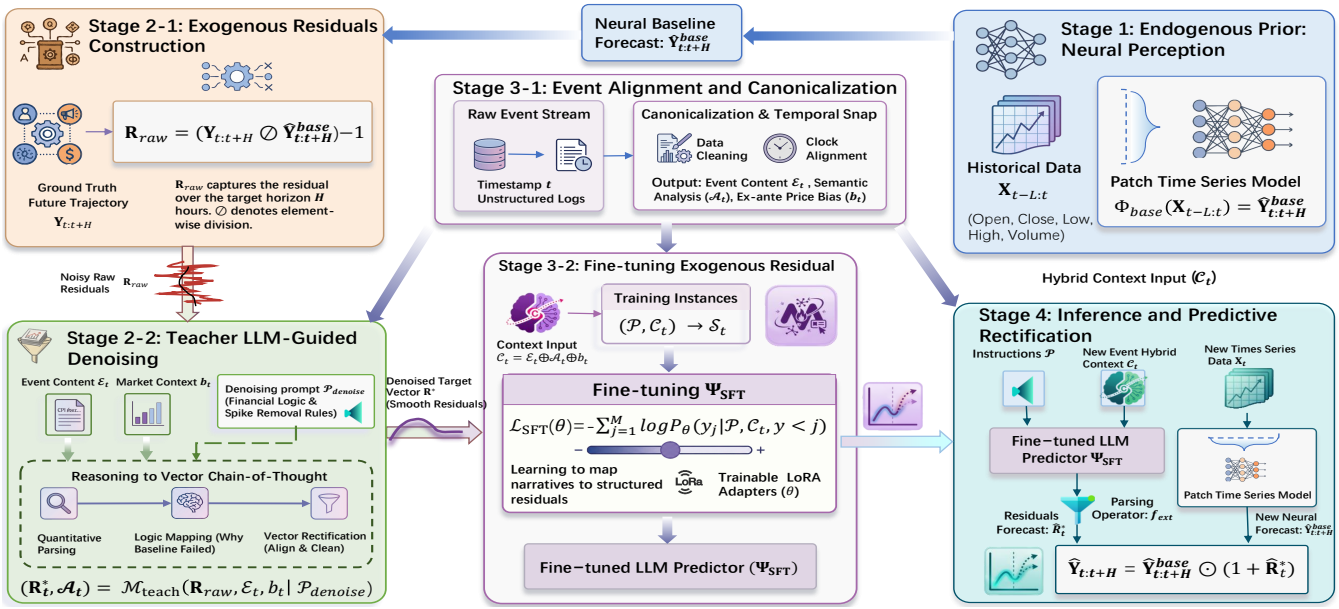

Figure 1: R2R: Reasoning-to-Residual Alignment Framework

whale orders). Training directly on raw residuals would force the LLM to hallucinate semantic reasons for random fluctuations.

To construct a high-quality instruction-tuning dataset, we implement a semantic denoising pipeline powered by an LLM (Gemini 3 Pro). As illustrated in Eq. 4, this process distills noisy market data into logically aligned correction vectors.

Stage 2-1: Raw Residual Calculation. For each macro event e_t, we first compute the *raw relative residual* $\mathbf{R}_{\text{raw}} \in \mathbb{R}^H$ by comparing the ground truth future trajectory $\mathbf{Y}_{t:t+H}$ with the endogenous-only baseline forecast:

$$\mathbf{R}_{\text{raw}} = (\mathbf{Y}_{t:t+H} \oslash \hat{\mathbf{Y}}^{\text{base}}_{t:t+H}) - \mathbf{1} \tag{3}$$

where $\oslash$ denotes element-wise division. This vector captures the total bias of the technical model over the target horizon H.

Stage 2-2: Teacher LLM-Guided Denoising. We employ the teacher LLM $\mathcal{M}_{\text{teach}}$ as a financial logic expert to filter $\mathbf{R}_{\text{raw}}$.

- **Teacher LLM Reasoning Input:** The model receives a composite input tuple consisting of the event content $\mathcal{E}_t$, the pre-event technical bias b_t, and the noisy raw residual $\mathbf{R}_{\text{raw}}$. The denoising prompt ($\mathcal{P}_{\text{denoise}}$) enforces a strict reasoning to vector chain-of-thought:
  - **Quantitative Parsing:** Describe the physical shift in data (e.g., CPI rose from 2.5% to 3.0%).
  - **Logic Mapping:** Explain why the technically-driven time series model failed to capture this shift (e.g., Model ignored inflationary pressure, leading to under-prediction of yields).
  - **Vector Rectification:** Generate a denoised target vector $\mathbf{R}^*$. The teacher LLM is explicitly instructed to: *Align the vector with the logic while cleaning random spikes.*
- **Teacher LLM Reasoning Output:** the teacher filters out aleatoric fluctuations and outputs:

$$(\mathbf{R}^*_t, \mathcal{A}_t) = \mathcal{M}_{\text{teach}}(\mathbf{R}_{\text{raw}}, \mathcal{E}_t, b_t \mid \mathcal{P}_{\text{denoise}}) \tag{4}$$

  where
  - **Semantic Analysis ($\mathcal{A}_t$):** The high-level logical parsing (e.g., Unemployment fell to 5.2%, signaling a strengthening labor market).
  - **Smooth Residuals ($\mathbf{R}^*_t \in \mathbb{R}^H$):** The vector of denoised smooth residuals.
  - **Reason**: A textual explanation distilled.

Finally, we discard the noisy $\mathbf{R}_{\text{raw}}$ and retain $\mathbf{R}^*$ as the smooth training target residual.

5.4 Stage 3: Fine-Tuning Exogenous Residual via Structured Generation

To bridge the modality gap between semantic policy events and numerical market corrections, we reformulate the residual prediction not as a simple regression task, but as a structured chain-of-thought generation problem. In this stage, the generic LLM is fine-tuned into a specialized semantic residual predictor, denoted as Ψ_{SFT}. Unlike traditional heads that output a single scalar, our model generates a comprehensive JSON object containing reasoning, discrete classification, and the continuous correction trajectory.

5.4.1 Stage 3-1: Event Alignment and Canonicalization. To ensure the fine-tuning process relies on strictly causal data, we implement a robust preprocessing pipeline before constructing the prompts.

Causal Synchronization and Safety Buffer. A primary challenge in financial modeling is the instantaneous reaction bias, where market prices often fluctuate the moment news is released. To ensure our system simulates a real-world predictive environment, we define a

causal safety buffer δ. For any policy event e released at time t_{release}, we synchronize it to a forecasting origin t_{origin} as follows:

$$t_{\text{origin}} = \lfloor t_{\text{release}} \rfloor_{\text{hour}} - \delta, \quad \delta = 1 \text{ hour} \tag{5}$$

By forcing the model to use the market state from at least one hour prior to the event, we guarantee that the input data contains no leaked information from the post-event volatility. This ensures that the LLM is performing a true ex-ante prediction rather than a retrospective reaction to realized price jumps.

Quantifying Ex-ante Price Bias. Even before a macro shock occurs, the technical backbone Φ_{base} typically exhibits a certain degree of forecasting error due to its inherent limitations. To provide the teacher LLM $\mathcal{M}_{\text{teach}}$ with this crucial baseline state, we calculate the ex-ante price bias b_t. We compute this by averaging the baseline model's relative errors over a $k = 4$ hour window immediately preceding t_{origin}:

$$b_t = \text{Mean}\left(\mathbf{Y}_{t-k:t} \oslash \hat{\mathbf{Y}}^{\text{base}}_{t-k:t} - \mathbf{1}\right) \tag{6}$$

This b_t acts as a calibration anchor. It informs the LLM whether the numerical model was already over-predicting or under-predicting before the news hit. By including this context, the teacher LLM $\mathcal{M}_{\text{teach}}$ can distinguish between the pre-existing model bias and the new shock caused by the macro event.

5.4.2 Training Instances Construction. The efficacy of Ψ_{SFT} depends on a prompt structure that integrates both semantic context and quantitative market state. Based on our aligned data, each training instance is defined as a tuple $(\mathcal{P}, \mathcal{C}, \mathcal{S})$, formatted as follows:

- **System Instruction ($\mathcal{P}$):** A persistent prompt conditioning the model to act as a macro-quant strategist. It explicitly defines the objective: *As a macro-aware layer for the time series model, predict the correction vector for the next H hours.*
- **Training Context Input (C_t):** To mitigate hallucination and provide the model with grounded state information, we define the hybrid context C_t as a multi-modal concatenation:

$$C_t = \mathcal{E}_t \oplus \mathcal{A}_t \oplus b_t \tag{7}$$

  where $\oplus$ denotes the string concatenation operator. The components are:
  - **Raw Event Content ($\mathcal{E}_t$):** The unstructured textual announcement aligned to t_{origin}.
  - **Semantic Analysis ($\mathcal{A}_t$):** The high-level logical parsing generated by the Teacher LLM ($\mathcal{M}_{\text{teach}}$) during the denoising stage (e.g., Unemployment fell to 5.2%, signaling a strengthening labor market).
  - **Ex-ante Price Bias (b_t):** As calculated in Eq. 6, this scalar provides the model with the current quantitative blind spot of the numerical encoder before the events impact.
- **Training Target Output ($\mathcal{S}_t$):** The ground-truth structured target is encoded as a JSON string. Each $\mathcal{S}_t$ contains:
  - `vector`: The vector of denoised smooth residuals $\mathbf{R}^*_t \in \mathbb{R}^H$ from the teacher LLM $\mathcal{M}_{\text{teach}}$.
  - `reason`: A textual explanation distilled from the Teacher LLM $\mathcal{M}_{\text{teach}}$.

5.4.3 Stage 3-2: Policy-Aware Parameter-Efficient Fine-Tuning. We employ LoRA to align the pre-trained LLM with the dynamics of financial residuals. The optimization objective is to train the model to approximate the mapping $(\mathcal{P}, C_t) \rightarrow \mathcal{S}_t$ by treating the structured string as a single coherent sequence.

Formally, let the ground-truth target string $\mathcal{S}_t$ be tokenized into a sequence $y_{1:M} = \{y_1, y_2, \ldots, y_M\}$, where M denotes the number of tokens. The model learns to minimize the causal language modeling (CLM) loss, calculated as the negative log-likelihood of predicting the next token y_j at position $j \in \{1, \ldots, M\}$ given the prefix $(\mathcal{P}, C_t)$ and all preceding tokens $y_{<j}$:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{j=1}^{M} \log P_\theta(y_j \mid \mathcal{P}, C_t, y_{<j}) \tag{8}$$

where θ denotes the learnable LoRA parameters. At inference time, the fine-tuned model Ψ_{SFT} generates the predicted structured response $\hat{\mathcal{S}}_t$ via autoregressive decoding. This generative approach allows the model to capture the non-linear dependencies between the qualitative reasoning tokens and the quantitative numerical tokens in the vector.

5.5 Stage 4: Synergistic Inference and Predictive Rectification

During the inference phase, R2R operates as a dual-stream system to produce a calibrated forecast, effectively combining the stability of numerical models with the adaptability of semantic reasoning.

5.5.1 Structured Generative Inference. During the inference phase, the fine-tuned predictor Ψ_{SFT} interprets the current macro-event stream to produce a high-fidelity correction. Given the instruction $\mathcal{P}$ and the synthesized context C_t, the inference process is formalized as:

$$\hat{\mathcal{S}}_t = \Psi_{\text{SFT}}(\mathcal{P}, C_t) \tag{9}$$

To translate the linguistic output into numerical signals, we define a parsing operator f_{ext} that isolates the `vector` from the generated JSON string:

$$\hat{\mathbf{R}}^*_t = f_{\text{ext}}(\hat{\mathcal{S}}_t.\text{vector}), \quad \hat{\mathbf{R}}^*_t \in \mathbb{R}^H \tag{10}$$

This design ensures that $\hat{\mathbf{R}}^*_t$ is a direct quantitative derivative of the preceding reasoning tokens within the JSON structure.

5.5.2 Multiplicative Fusion. The final forecast $\hat{\mathbf{Y}}_{t:t+H}$ is produced by rectifying the endogenous technical prior with the semantic residual. The synthesis is governed by the following multiplicative modulation:

$$\hat{\mathbf{Y}}_{t:t+H} = \hat{\mathbf{Y}}^{\text{base}}_{t:t+H} \odot (\mathbf{1} + \hat{\mathbf{R}}^*_t) \tag{11}$$

This architecture ensures that the LLM acts as a dynamic scaler. In the absence of significant news, Ψ_{SFT} tends to predict a zero-vector, causing the term $(1 + \hat{\mathbf{R}}^*_t)$ to approach identity, thereby preserving the high-fidelity technical trend. Conversely, during high-impact shocks, the LLM injects a non-linear correction, effectively bending the technical curve to align with the new macro regime.

Table 1: Horizon-specific forecasting performance on seven assets (BTC, ETH, DOGE, SOLANA, SPX, NDX, INDU) using hourly (1h) OHLCV data. Forecast horizons include t+1h, t+2h, t+6h, t+12h, and t+24h. Best in red; second-best in blue.

Asset	Horizon	R2R (Ours)			Time-LLM [6]			GPT4TS [30]			PatchTST [15]			iTransformer [10]			DLinear [25]		
		MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE
BTC	*Overall*	**0.000280**	**0.011980**	**1.2380**	0.000428	0.015567	1.5923	0.000303	0.012550	1.2973	0.000369	0.014224	1.4690	0.000381	0.014277	1.4779	0.000551	0.018398	1.8896
	$t+1$	**0.000052**	**0.005073**	**0.5249**	0.000130	0.008249	0.8480	0.000054	0.005190	0.5376	0.000063	0.005736	0.5882	0.000081	0.006556	0.6770	0.000198	0.011139	1.1357
	$t+2$	**0.000089**	**0.006757**	**0.7012**	0.000183	0.010104	1.0363	0.000092	0.006916	0.7177	0.000113	0.007723	0.7938	0.000134	0.008539	0.8821	0.000244	0.012324	1.2455
	$t+6$	**0.000205**	**0.010601**	**1.1048**	0.000385	0.015600	1.6055	0.000216	0.010996	1.1467	0.000276	0.012781	1.3230	0.000292	0.013026	1.3550	0.000439	0.017036	1.7438
	$t+12$	**0.000283**	**0.012513**	**1.2911**	0.000411	0.015464	1.5784	0.000306	0.013165	1.3589	0.000364	0.014804	1.5294	0.000384	0.014824	1.5328	0.000519	0.018162	1.8758
	$t+24$	**0.000392**	**0.014899**	**1.5301**	0.000570	0.018445	1.8767	0.000456	0.016216	1.6555	0.000576	0.018562	1.9110	0.000567	0.018399	1.8912	0.000747	0.021457	2.2047
ETH	*Overall*	**0.000768**	**0.019722**	**2.0168**	0.001307	0.026672	2.6821	0.000933	0.021997	2.2331	0.000951	0.022203	2.2155	0.001027	0.023238	2.3252	0.001528	0.029438	2.9317
	$t+1$	0.000171	0.008482	**0.8368**	0.000408	0.013926	1.4102	0.000183	0.008876	0.8734	**0.000158**	**0.008474**	0.8539	0.000203	0.009666	0.9744	0.000548	0.017757	1.7917
	$t+2$	**0.000247**	**0.011076**	**1.1123**	0.000543	0.016956	1.7098	0.000282	0.011885	1.1836	0.000267	0.011791	1.1770	0.000308	0.012629	1.2676	0.000678	0.019841	1.9532
	$t+6$	**0.000597**	**0.017153**	**1.7468**	0.001154	0.025923	2.6149	0.000726	0.019411	1.9589	0.000680	0.019067	1.8994	0.000714	0.019690	1.9838	0.001193	0.025866	2.5709
	$t+12$	**0.000741**	**0.019723**	**2.0260**	0.001244	0.026906	2.7335	0.000902	0.022191	2.2733	0.000898	0.022102	2.2254	0.000966	0.023457	2.3719	0.001397	0.028659	2.8808
	$t+24$	**0.001357**	**0.028191**	2.8984	0.001891	0.032438	3.1804	0.001399	0.028484	**2.8650**	0.001551	0.029798	2.9422	0.001684	0.031312	3.1111	0.002106	0.035234	3.4460
SOL	*Overall*	**0.001064**	**0.024373**	**2.4828**	0.001583	0.031359	3.1371	0.001263	0.027261	2.7818	0.001258	0.026761	2.6903	0.001330	0.027896	2.8349	0.002045	0.035654	3.5508
	$t+1$	0.000222	0.010826	1.0979	0.000457	0.016081	1.6229	0.000231	0.011272	1.1401	**0.000195**	**0.010109**	**1.0243**	0.000303	0.013044	1.3364	0.000820	0.022735	2.2778
	$t+2$	0.000315	0.012841	1.3178	0.000592	0.019040	1.9258	0.000345	0.013427	1.3732	**0.000299**	**0.012736**	**1.2860**	0.000406	0.015326	1.5683	0.000906	0.024213	2.3930
	$t+6$	**0.000731**	**0.020672**	**2.1208**	0.001212	0.028210	2.8456	0.000827	0.022521	2.3065	0.000748	0.021253	2.1453	0.000801	0.022173	2.2654	0.001432	0.030701	3.0548
	$t+12$	**0.001065**	**0.025368**	**2.5854**	0.001567	0.032106	3.2295	0.001254	0.028336	2.8937	0.001197	0.027315	2.7521	0.001272	0.028277	2.8783	0.001906	0.035606	3.5634
	$t+24$	**0.001509**	**0.030960**	**3.1307**	0.002284	0.038303	3.7826	0.001961	0.035795	3.6113	0.002097	0.036430	3.6175	0.002193	0.037396	3.7542	0.002858	0.042667	4.2123
DOGE	*Overall*	**0.001359**	**0.026047**	**2.5887**	0.001955	0.031832	3.1204	0.001566	0.029348	2.9153	0.002434	0.031263	2.9485	0.002721	0.032276	3.0602	0.003183	0.038343	3.7145
	$t+1$	**0.000235**	**0.010660**	1.0580	0.000539	0.016136	1.5891	0.000320	0.013048	1.3106	0.000540	0.011157	**1.0114**	0.000712	0.013637	1.2518	0.001337	0.023893	2.3137
	$t+2$	**0.000331**	**0.013363**	**1.3433**	0.000664	0.018707	1.8542	0.000441	0.015992	1.6101	0.000856	0.015893	1.5020	0.000965	0.016682	1.5611	0.001520	0.024797	2.2744
	$t+6$	**0.000741**	**0.020839**	**2.0943**	0.001343	0.027431	2.7084	0.000889	0.023180	2.3247	0.001465	0.023450	2.2428	0.001427	0.023450	2.2704	0.002251	0.032244	3.1381
	$t+12$	**0.001252**	**0.026353**	**2.5982**	0.001822	0.032292	3.1676	0.001421	0.029581	2.9421	0.002272	0.031299	2.9921	0.002582	0.033213	3.1739	0.002935	0.037266	3.5953
	$t+24$	**0.002518**	**0.036224**	**3.5976**	0.003254	0.041538	4.0223	0.002604	0.039516	3.9011	0.004236	0.043718	4.1291	0.004503	0.043944	4.1741	0.004474	0.044169	4.1171
SPX	*Overall*	**0.000084**	**0.005783**	**0.6054**	0.000124	0.007218	0.7531	0.000092	0.006253	0.6525	0.000261	0.010526	1.1035	0.000260	0.010422	1.0937	0.000331	0.012485	1.3036
	$t+1$	**0.000043**	**0.004097**	**0.4272**	0.000089	0.006109	0.6390	0.000046	0.004355	0.4534	0.000062	0.004942	0.5189	0.000066	0.005188	0.5427	0.000125	0.007757	0.8086
	$t+2$	**0.000054**	**0.004652**	**0.4855**	0.000098	0.006324	0.6591	0.000058	0.004881	0.5079	0.000078	0.005480	0.5721	0.000081	0.005815	0.6069	0.000135	0.007956	0.8265
	$t+4$	**0.000080**	**0.005682**	**0.5945**	0.000106	0.006818	0.7098	0.000085	0.006042	0.6306	0.000086	0.006147	0.6416	0.000100	0.006746	0.7019	0.000169	0.010033	1.0285
	$t+6$	**0.000100**	**0.006269**	**0.6571**	0.000128	0.007363	0.7676	0.000110	0.006850	0.7151	0.000129	0.007675	0.7977	0.000115	0.006879	0.7184	0.000171	0.009111	0.9401
	$t+8$	**0.000150**	**0.007900**	**0.8328**	0.000211	0.009352	0.9826	0.000165	0.008674	0.9107	0.000198	0.009469	0.9911	0.000190	0.009129	0.9597	0.000261	0.010914	1.1391
NDX	*Overall*	**0.000121**	**0.007182**	**0.7568**	0.000167	0.009007	0.9466	0.000127	0.007584	0.7986	0.000323	0.012605	1.3237	0.000332	0.012829	1.3485	0.000492	0.016113	1.6926
	$t+1$	**0.000061**	**0.005010**	**0.5286**	0.000120	0.007607	0.8044	0.000063	0.005237	0.5521	0.000070	0.005711	0.5995	0.000094	0.006838	0.7139	0.000198	0.009977	1.0492
	$t+2$	**0.000076**	**0.005740**	**0.6048**	0.000131	0.007483	0.7842	0.000082	0.006065	0.6381	0.000093	0.006558	0.6884	0.000112	0.007360	0.7697	0.000210	0.010185	1.0668
	$t+4$	**0.000113**	**0.007103**	**0.7501**	0.000151	0.008870	0.9301	0.000120	0.007545	0.7961	0.000127	0.007985	0.8394	0.000139	0.008285	0.8706	0.000270	0.012847	1.3260
	$t+6$	**0.000144**	**0.007831**	**0.8255**	0.000181	0.009290	0.9752	0.000149	0.008198	0.8645	0.000153	0.008403	0.8830	0.000152	0.008463	0.8910	0.000270	0.011790	1.2255
	$t+8$	**0.000212**	**0.009787**	**1.0345**	0.000265	0.011670	1.2313	0.000221	0.010461	1.1060	0.000233	0.010875	1.1445	0.000234	0.010908	1.1509	0.000386	0.014252	1.4981
INDU	*Overall*	**0.000078**	**0.005439**	**0.5573**	0.000101	0.006472	0.6624	0.000079	0.005639	0.5777	0.000206	0.009158	0.9432	0.000217	0.009496	0.9767	0.000273	0.011568	1.1854
	$t+1$	**0.000038**	**0.003635**	**0.3743**	0.000069	0.005252	0.5384	0.000039	0.003806	0.3916	0.000041	0.004072	0.4185	0.000051	0.004476	0.4593	0.000093	0.006760	0.6918
	$t+2$	**0.000053**	**0.004410**	**0.4525**	0.000078	0.005586	0.5715	0.000055	0.004542	0.4661	0.000053	0.004831	0.4944	0.000064	0.005062	0.5171	0.000108	0.007255	0.7407
	$t+4$	0.000084	**0.005423**	**0.5553**	0.000103	0.006332	0.6481	0.000084	0.005544	0.5674	**0.000074**	0.005485	0.5620	0.000089	0.006077	0.6214	0.000145	0.008842	0.8970
	$t+6$	0.000092	**0.005952**	**0.6087**	0.000110	0.006999	0.7151	0.000095	0.006267	0.6412	**0.000088**	0.006448	0.6590	0.000098	0.006574	0.6712	0.000138	0.008666	0.8809
	$t+8$	0.000104	**0.006940**	**0.7116**	0.000129	0.007732	0.7919	0.000107	0.007166	0.7346	**0.000102**	0.007151	0.7321	0.000117	0.007518	0.7697	0.000159	0.009177	0.9354

6 Experiments

We conduct extensive experiments to validate the effectiveness, robustness, and practicality of the proposed R2R framework. Our evaluation is organized around the following research questions:

- **RQ1 (Overall Performance):** Does R2R outperform state-of-the-art LTSF models and LLM-based forecasting methods?
- **RQ2 (Mechanism Validity):** What are the contributions of instruction tuning and residual quantization/injection design?
- **RQ3 (Policy Impact Analysis):** What performance gains does R2R achieve when incorporating different types of macro-policy signals?
- **RQ4 (Practical Utility):** Do forecasting gains translate into improved trading performance and risk-adjusted returns?
- **RQ5 (Efficiency):** What is the computational overhead of R2R compared with lightweight forecasters and LLM-based baselines?
- **PQ6 (Case Study):** Forecast correction under policy events

6.1 Experimental Setup

6.1.1 Datasets and Benchmarks. We use hourly market data from Jan 1, 2021, to Dec 31, 2025, covering multiple volatility regimes. We evaluate across seven assets spanning both crypto (BTC, ETH, DOGE, SOLANA) and equity markets (SPX, NDX, INDU) using historical market data from Binance and Yahoo Finance [2, 23].

Data split. We adopt a strictly chronological split to prevent look-ahead bias. Specifically, the full time range is divided by time order into training/validation/test with a 7:1:2 ratio (from earlier

Table 2: Ablation study on ETH ($H = 24$). Δ is improv. over GPT4TS.

Configuration / Variant	MSE	MAE	MAPE (%)	Δ MAE Improv.	Δ MSE Improv.
Standard Technical Baseline					
(1) GPT4TS Only (Base)	0.000933	0.021997	2.2331	-	-
(2) PatchTST Only (Base)	0.000951	0.022203	2.2155	-	-
(1) Gemini-v3 Only (Base)	0.000711	0.019218	1.9568	-	-
(2) GPT-V5 Only (Base)	0.000804	0.020147	2.0472	-	-
Semantic Integration Variants					
(3) w/o Instruction-Tuning (Zero-Shot)	0.000917	0.023078	2.3548	-4.92%	+1.73%
(2) w/o Real Residual Injection	0.000850	0.020738	2.1117	+5.72%	+8.92%
(2) w/o Residual Injection (Early Fusion)	0.0009	0.0219	2.2194	+0.44%	+3.53%
(2) w/o Smooth Residual Injection	0.000821	0.020358	2.0813	+7.45%	+12.05%
(4) w/o ex-ante price bias Injection	0.000768	0.019722	2.0168	+10.34%	+17.66%
LLM Backbone Sensitivity					
(5a) Qwen-2.5-1.5B + R2R	0.000680	0.019024	1.9382	+13.51%	+21.10%
(5b) Qwen-2.5-7B + R2R	0.000704	0.019365	2.22	+11.97%	+24.54%
(5c) Llama-3-8B + R2R	0.000766	0.019827	2.0182	+9.86%	+17.95%
(5d) Qwen-3-8B + R2R	0.000768	0.019722	2.0168	+10.34%	+17.66%
Full R2R Framework	0.000768	0.019722	2.0168	+10.34%	+17.66%

to later), i.e., the earliest 70% for training, the subsequent 10% for validation, and the most recent 20% for testing.

Macro-event stream. We construct a macro-policy event dataset aligned with the market timeline to evaluate one-day-ahead prediction improvement after policy releases. The dataset includes major announcements (see Section 3), each timestamped and synchronized with hourly market data. Using the same chronological split, the policy samples include 1150 training, 168 validation, and 311 test events, with split boundaries at Sep 25, 2024, 13:00 and Feb 27, 2025, 12:00.

We compare against two groups of baselines: (1) **SOTA time-series models:** GPT4TS [30] (also used as a backbone), PatchTST [15], iTransformer [10], and DLinear [25]. (2) **LLM-based methods:** Time-LLM [6] and generic LLM baselines that concatenate textual embeddings with time-series features.

6.1.2 Implementation Details. We adopt a look-back window $L = 96$ and evaluate multiple horizons $H \in \{1, \ldots, 24\}$ for forecasting. For the semantic stream, we use Qwen-3-8B as the base model and apply LoRA fine-tuning on the training split with rank $r = 16$ and $\alpha = 32$. All evaluations are strictly chronological to avoid leakage. We report MSE/MAE/MAPE for forecasting, and additionally CR/SR/MDD/DA for trading.

6.2 Main Results and Analysis (RQ1)

Table 1 reports forecasting errors across seven assets. R2R achieves the best or second-best results in nearly all settings, demonstrating strong and consistent superiority over competitive baselines, with particularly strong advantages in long-horizon and high-volatility settings, validating the effectiveness of the proposed framework.

Clear overall improvement. R2R consistently outperforms the strongest baselines, reducing Overall MSE by **7.6%** (BTC), **17.7%** (ETH), **15.4%** (SOL), and **13.2%** (DOGE), with additional gains across equity indices. On average, R2R achieves about **8.0%** improvement, indicating more stable and reliable predictions.

Larger gains at long horizons. The advantage becomes more pronounced as the horizon increases. For example, on BTC at t+24, R2R reduces MSE by **14.0%** and MAE by **8.1%** versus the best competing model, confirming stronger robustness under regime shifts.

Table 3: Impact of Top 10 Policy Events on BTC Volatility: MAE Reduction (%) at horizons 12H, 18H, and 24H.

Asset	Horizon	Policy Events									
		Proj	Powell	Retail	Home	Bond	Crude	C.CPI	C.PCE	GDP	ISM
BTC	Overall	12.06	12.39	8.17	7.19	5.03	5.77	6.46	5.17	5.24	5.51
	T+12h	12.71	10.33	2.18	6.05	6.55	5.86	5.04	6.82	6.78	7.04
	T+18h	17.89	15.85	6.08	6.21	5.62	7.40	15.15	7.02	9.88	3.82
	T+24h	8.07	18.29	33.64	17.88	16.40	11.57	4.66	18.37	7.81	3.56
ETH	Overall	13.18	31.30	15.85	12.75	8.79	13.40	1.27	9.80	27.65	13.01
	T+1h	-10.03	75.55	-10.10	-14.85	-17.43	6.96	-23.39	-5.48	43.50	5.61
	T+15h	14.43	40.39	19.68	1.25	9.04	19.28	3.27	9.63	47.43	17.91
	T+23h	27.18	11.79	48.60	16.58	25.14	12.70	-24.89	8.54	32.07	12.32
DOGE	Overall	21.11	33.32	22.77	0.47	4.69	17.10	-17.39	19.61	10.60	25.08
	T+1h	45.19	8.73	33.32	19.82	3.47	12.75	24.56	31.77	28.66	20.13
	T+14h	26.41	30.14	31.58	-2.89	3.60	19.63	-2.13	15.62	9.78	28.01
	T+20h	42.74	57.50	30.82	-7.83	4.11	24.99	-16.99	21.70	3.84	23.99
SOL	Overall	12.32	2.43	27.52	12.63	8.53	11.44	13.48	13.51	3.85	8.39
	T+4h	12.46	1.05	16.29	8.83	4.51	4.63	12.22	8.06	6.24	22.61
	T+20h	0.46	1.04	30.23	11.10	12.20	15.95	16.54	20.04	5.45	4.67
	T+24h	38.32	29.44	50.45	17.65	10.71	19.76	30.93	29.79	2.55	0.43
SPX	Overall	16.35	10.67	4.61	4.73	5.00	5.64	-3.14	9.55	-0.98	5.07
	T+1h	-0.40	8.87	7.35	12.15	-5.15	6.29	3.09	11.57	2.60	7.33
	T+5h	25.72	-0.23	2.98	4.39	16.49	6.99	-3.77	8.60	0.62	-0.36
	T+8h	39.93	23.61	11.83	0.45	9.62	5.42	-7.12	9.08	-1.70	7.49
NDX	Overall	14.94	23.23	-15.33	-9.75	-0.78	2.64	11.01	9.42	12.12	-2.67
	T+2h	27.46	29.19	-6.33	-8.54	-19.36	6.51	15.45	9.19	40.73	-5.73
	T+4h	10.30	22.86	-14.95	-1.01	8.34	-1.53	11.81	12.13	15.15	-2.24
	T+7h	19.91	13.28	-25.23	-3.73	14.80	3.29	7.01	7.46	7.98	1.55
INDU	Overall	12.63	15.28	13.11	-3.23	4.98	2.01	-3.46	-3.32	4.97	4.28
	T+1h	3.12	10.56	17.84	0.82	-4.21	3.07	0.58	4.51	6.42	3.91
	T+6h	24.99	24.68	7.63	-5.30	16.41	0.20	-9.02	-5.84	17.48	7.11
	T+7h	23.54	25.43	6.14	-7.94	12.80	3.18	-9.12	-4.42	18.25	8.76

6.3 Ablation Study (RQ2)

We conduct a step-wise ablation that progressively removes modules from the full R2R pipeline, matching its dependency chain. This ensures each removal reflects its true functional role rather than an isolated tweak.

Starting from full R2R, we remove components in order: (i) **ex-ante bias & smoothing**, (ii) **real residual injection**, (iii) **late injection → early fusion**, (iv) **SFT (zero-shot)**, and finally (v) **pure technical baselines**. Table 2 shows a **monotonic performance drop** along this sequence: full R2R achieves **+10.34% MAE** and **+17.66% MSE** over GPT4TS, while removing real residual injection nearly halves the gain (**+5.72% MAE**); replacing late injection with early fusion collapses the benefit (**+0.44% MAE**); and removing SFT further degrades to **-4.92% MAE**. This step-wise degradation confirms that each module is necessary and complementary, and that the final improvement arises from their **joint effect** rather than any single component.

6.4 Policy Event Impact Analysis (RQ3)

Table 3 shows that adding structured macro-policy signals consistently reduces MAE across assets and horizons, with the strongest gains in crypto. BTC improves most under Powell (+12.39%) and Project (+12.06%), while ETH and DOGE achieve larger gains, including Powell (+31.30%) and GDP (+27.65%) for ETH, and Powell

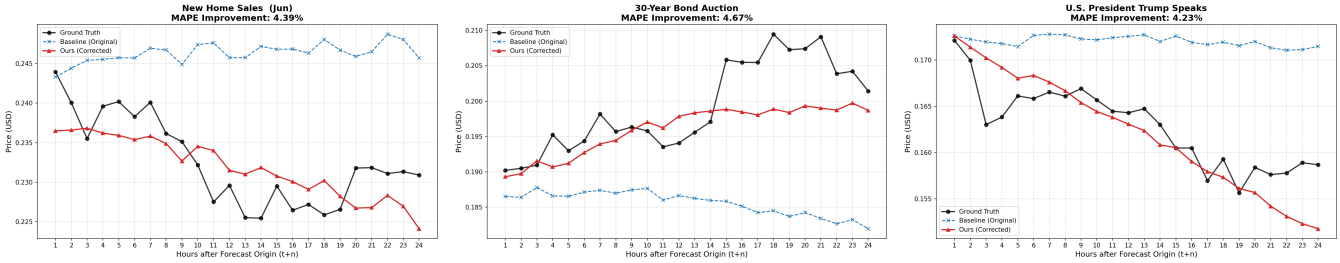

Figure 2: Forecast Correction under Policy Events

(+33.32%) for DOGE, confirming the framework captures policy-driven volatility beyond purely statistical models.

Policy effects are horizon-dependent and cross-asset, with larger gains at longer horizons (e.g., BTC Retail +33.64%, SOL Retail +50.45% at 24h), indicating persistent macro influence. Significant improvements also appear in equities (SPX +39.93%, NDX +40.73%), showing cross-market generalization, with some pairs reaching very large gains (e.g., ETH under Powell +75.55%). Overall, consistent and economically interpretable improvements demonstrate that policy-event integration substantially enhances volatility forecasting and validates the proposed framework.

6.5 Risk-Adjusted Trading Performance (RQ4)

We assess practical value via a trading simulation on BTC and INDU with horizon $H = 24$ (Table 4). R2R delivers the most stable risk-adjusted performance across both markets and is the only model with a non-negative Sharpe ratio on BTC (0.0153), while all baselines remain negative. On INDU, R2R achieves the highest cumulative return (35.02%) and highest Sharpe ratio (2.1660), demonstrating superior profitability and consistency.

In the volatile BTC market, R2R reduces loss to -2.74% (vs. -14.66% PatchTST and -33.26% GPT4TS) with controlled drawdown, indicating stronger robustness. On INDU, it again outperforms all baselines in both return and Sharpe ratio. As directional accuracy is similar across models, the gains mainly stem from better magnitude estimation and risk-aware corrections, showing that R2R effectively converts predictive improvements into consistent financial gains.

Table 4: Trading performance on BTC and INDU ($H = 24$).

Model	Crypto (BTC)				Equity (INDU)			
	CR ↑	SR ↑	MDD ↓	DA ↑	CR ↑	SR ↑	MDD ↓	DA ↑
Buy-and-Hold	-73.96%	-2.3491	-82.59%	-	-11.04%	-0.3034	-44.21%	-
PatchTST	-14.66%	-0.4720	-50.96%	50.87%	28.21%	1.8166	-12.43%	47.76%
GPT4TS	-33.26%	-1.4667	-53.22%	51.41%	27.05%	1.7975	-10.01%	49.78%
Time-LLM	-15.25%	-0.6119	-42.03%	50.80%	30.03%	1.9747	-9.79%	48.29%
R2R (Ours)	**-2.74%**	**0.0153**	**-48.97%**	**50.87%**	**35.02%**	**2.1660**	**-11.85%**	**48.26%**

6.6 Efficiency Analysis (RQ5)

Table 5 compares parameter size and inference latency across forecasting and LLM-based methods. Despite using a large-model backbone, R2R achieves **competitive latency (4681 ms)** due to its decoupled neuro-symbolic design, where semantic reasoning is invoked mainly during policy-event windows and introduces minimal overhead otherwise.

Table 5: Latency is the average inference time per step (ms).

Metric	DLinear	iTransformer	PatchTST	GPT4TS	Time-LLM	Gemini-v3	GPT-v5	R2R (Ours)
Parameters	0.02 M	0.15 M	0.45 M	0.45 M	8.1 B	8.1 B	8.1 B	**8.1 B**
Latency (ms)	6.2	11.5	12.8	507.5	3845.0	8820.0	7460.3	**4681.0**

*Latency measured during event windows; negligible overhead during normal periods.

Compared with other LLM-based approaches, R2R reduces latency by **46.9% vs. Gemini-v3**, **37.3% vs. GPT-v5**, and **17.0% vs. Time-LLM** at a comparable parameter scale. Although lightweight models (e.g., DLinear, PatchTST) are faster, they lack reasoning capability and perform substantially worse in forecasting and trading tasks. Overall, R2R delivers a favorable **accuracy–efficiency trade-off**, enabling strong performance with controlled computational cost for practical deployment.

6.7 Case Study: Policy-Aware Correction (RQ6)

We present three representative policy-event cases to qualitatively evaluate the effectiveness of the proposed residual correction. As shown in Fig. 2, the corrected forecasts consistently align more closely with the true price trajectory than the baseline. The correction reduces systematic drift after the 30-Year Bond Auction (MAPE +4.67%), captures the post-event downward regime in New Home Sales (MAPE +4.39%), and adjusts trajectory direction following a major presidential speech (MAPE +4.23%). These results demonstrate that the proposed framework effectively mitigates forecast bias under macro-policy-driven regime shifts.

7 Limitations

Despite its efficacy, the R2R framework has several constraints. First, the **generative regression** approach relies on numerical tokenization, which may introduce quantization noise compared to traditional linear regression heads. Second, the framework is **event-triggered**, meaning its superior performance is concentrated during defined macro-economic windows, while its ability to capture non-narrative-driven volatility remains identical to the underlying numerical backbone.

8 Conclusion

R2R integrates macro-policy to improve volatility forecasting. Results show consistent accuracy gains and stronger performance, demonstrating robust adaptation to diverse macro-policy conditions.

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A Prompts

Denoising Prompt $\mathcal{P}_{\text{denoise}}$

Role: Financial Data Logic Expert.

Context: You are analyzing the prediction errors of a time-series model that ONLY looks at historical price patterns and is blind to macro-policy events. The provided raw residuals represent the model's prediction bias caused by its ignorance of the specific macro event.

Task:

1. Quantitative Analysis: Compare 'Actual' vs 'Previous'. Describe the physical change (no more than 20 words).
2. Logic Mapping: Explain why the blind Patch timeseries model failed to capture the raw residuals given this macro shift (no more than 20 words).
3. `correction_vector`: Your predicted H hours denoised version of Ground Truth. Align with logic while cleaning random spikes.

User Input Template (Dynamic Context)

`— Macro Event —`

Event: `{event_name}`

Data: Actual: `{actual}`, Previous: `{previous}`, Forecast: `{forecast}`

`— Statistics —`

4h Historical Drift: `{drift_4h}`%

Max Detected Offset in GT: `{max_offset}`%

`— Target (Time-series model's Errors) —`

Ground Truth Residuals (t_{+1} to t_{+H}): `[v1, v2, ..., vH]`

Instruction: Generate the data_description and logic_explanation and distilled_vector based on macro logic.

Output Format (Strict JSON):

```
{
  "data_description": "...",
  "logic_explanation": "...",
  "correction_vector": [H floats]
}
```

Figure 3: The system prompt for the Teacher LLM (Denoising Agent). It integrates the Role, Task, and dynamic User Input (Macro Event & Residuals) to distill deterministic policy signals, ensuring the student model learns grounded economic logic.

Fine-tuning Data Sample Construction

Role: Senior Macro-Quant Strategist.

Context: You act as a macro-aware correction layer for GPT4TS, a time-series model blind to external events. You are provided with a Macro Event Analysis and the 'Current Bias' (average prediction error over the past 4 hours, where positive = under-predicting).

Task:

1. INTEGRATION: Combine the macro shock analysis with the historical drift (Current Bias).
2. PREDICTION: Generate a 24-hour percentage correction vector for hourly prices.
3. FORMATTING: Output a structured JSON containing concise logic and the correction array.

User Input Template (Dynamic Context)

`— Input Data —`

Event: `{event_name}`

Analysis: `{macro_analysis_text}`

Current Bias: `{bias_value}%`

Output Format (Strict JSON):

```
{
  "reasoning": "Concise logic integrating macro event (max 20 words)",
  "vector": [24 floats representing % correction]
}
```

Figure 4: The format of a single sample in the fine-tuning dataset. Each instance pairs a specific macro scenario and model bias (Input) with the ground-truth reasoning and correction vector (Target), supervising the LLM to learn the mapping from semantic news to quantitative adjustments.

Ablation Data Sample Construction (w/o Historical Drift)

Role: Senior Macro-Quant Strategist.

Context: You act as a macro-aware correction layer for GPT4TS, a time-series model blind to external events. You are provided with a Macro Event Analysis.

Task:

1. ANALYSIS: Interpret the macro shock based on the provided event data.
2. PREDICTION: Generate a 24-hour percentage correction vector based **solely** on the macro data.
3. FORMATTING: Output a structured JSON containing concise logic and the correction array.

User Input Template (Dynamic Context)

`— Input Data —`

Event: `{event_name}`

Analysis: `{macro_analysis_text}`

Output Format (Strict JSON):

```
{
  "reasoning": "Concise macro logic (max 20 words)",
  "vector": [24 floats representing % correction]
}
```

Figure 5: The data sample format for the ablation study. Compared to the full method, this variant removes the Current Bias (historical drift) from the input, forcing the LLM to predict the correction vector relying exclusively on the semantic information of the macro event.